CHAPTER 9

HYDROCARBONS

CHAPTER ESSENCE AT A GLANCE:
Studies alkanes, alkenes, alkynes, and arenas; ethane conformations, Markovnikov additions, ozonolysis, and electrophilic aromatic substitution.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Ethane Conformations:** Rotation about C-C single bond yields staggered (most stable, minimum torsional strain) and eclipsed (least stable) forms.

• **Markovnikov Addition & Peroxide Effect:** HX adds to asymmetrical alkenes placing negative part on less hydrogenated carbon; peroxide reverses addition for HBr only.

• **Ozonolysis Cleavage:** Alkenes react with ozone followed by Zn with H2O reductive cleavage to give carbonyl products, revealing double bond position.

• **Huckel's Aromaticity Rule:** Cyclic, planar, fully conjugated ring systems possessing (4n + 2) pi electrons display exceptional aromatic stability.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$CH_3CH{=}CH_2 + HBr \rightarrow CH_3\text{-}CH(Br)\text{-}CH_3 \quad (\text{Markovnikov Product})$$

Markovnikov Electrophilic Addition to Propene

$$CH_3CH{=}CH_2 + HBr \xrightarrow{\text{peroxide}} CH_3CH_2CH_2Br \quad (\text{Anti-Markovnikov})$$

Kharasch Peroxide Effect (Specific to HBr)

$$(4n+2)\,\pi \text{ electrons} \quad (n = 1 \rightarrow 6\pi \text{ in Benzene})$$

Hückel's Rule of Aromaticity

$$2\text{R-X} + 2\text{Na} \xrightarrow{\text{dry ether}} \text{R-R} + 2\text{NaX}$$

Wurtz Reaction for Symmetrical Alkane Synthesis

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** The peroxide effect (anti-Markovnikov addition) works ONLY with HBr; it does NOT occur with HCl or HI because of thermodynamic bond enthalpies!

• **Alert:** Benzene resists addition reactions and prefers electrophilic substitution because substitution preserves the 36 kcal/mol aromatic resonance stabilization.

• **Alert:** Staggered conformation of ethane is 12.5 kJ/mol more stable than eclipsed due to zero torsional strain.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State Huckel's rule of aromaticity. Is the cyclopentadienyl anion aromatic? Justify your answer. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Huckel's Rule:** Any cyclic, planar, completely conjugated ring system containing (4n + 2) pi electrons (where n = 0, 1, 2, 3...) possesses exceptional stability and is aromatic. **[1 Mark]**

• **Cyclopentadienyl Anion Justification:** Cyclopentadienyl anion is cyclic, planar, and fully conjugated with 2 double bonds (4 pi electrons) plus 1 lone pair on carbon, totaling 6 pi electrons. For 4n + 2 = 6, n = 1 (an integer). Hence it is aromatic. **[1 Mark]**

• **Key Exam Result:** Cyclopentadienyl anion possesses 6 pi electrons obeying Huckel's rule (n=1) and is aromatic.

2. Describe the staggered and eclipsed conformations of ethane. Which conformation is more stable and why? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Conformations:** Rotation around the C-C single bond yields: (1) Eclipsed conformation where C-H bonds on adjacent carbons align directly behind one another with dihedral angle 0 degrees. (2) Staggered conformation where C-H bonds are maximally separated with dihedral angle 60 degrees. **[1 Mark]**

• **Stability Reasoning:** Staggered conformation is more stable (by 12.55 kJ/mol) because electron clouds of C-H bonds are at maximum distance, minimizing torsional strain and electron-electron repulsion. **[1 Mark]**

• **Key Exam Result:** Staggered conformation has minimum torsional strain and maximum thermodynamic stability.

3. State Markovnikov's rule. Explain the electrophilic addition mechanism of HBr to propene. What is the Peroxide Effect? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Markovnikov's Rule:** In the addition of an unsymmetrical reagent (HX) to an unsymmetrical alkene, the negative part of the addendum attaches to that double-bonded carbon which holds fewer hydrogen atoms. **[1 Mark]**

• **Mechanism:** H+ adds to propene forming a 2° carbocation (CH3-CH(+)-CH3) which is more stable than 1° carbocation. Subsequent fast attack by Br- gives 2-bromopropane as major product. **[1 Mark]**

• **Peroxide Effect (Kharasch):** In the presence of organic peroxides, addition of HBr proceeds via free-radical mechanism yielding 1-bromopropane (Anti-Markovnikov). Works ONLY with HBr. **[1 Mark]**

• **Key Exam Result:** Markovnikov major product = 2-bromopropane; Peroxide major product = 1-bromopropane.

$$\mathrm{CH_3CH{=}CH_2 + HBr \rightarrow CH_3\text{-}CH(Br)\text{-}CH_3 \quad (via\ 2^\circ\ carbocation)}$$

Electrophilic Addition to Propene

4. Explain ozonolysis of alkenes. An alkene 'A' on ozonolysis followed by cleavage with Zn and H2O yields ethanal and propanone. Deduce the structure and IUPAC name of 'A'. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Principle of Ozonolysis:** O3 adds across the double bond forming a cyclic ozonide, which undergoes reductive cleavage with Zn and H2O to break the C=C bond and yield carbonyl compounds (aldehydes or ketones). **[1 Mark]**

• **Deduction of Alkene 'A':** Products are Ethanal (CH3-CH=O) and Propanone (O=C(CH3)2). Removing oxygen atoms and joining the carbonyl carbon atoms with a double bond gives: CH3-CH=C(CH3)2. **[1 Mark]**

• **IUPAC Name:** 4-carbon chain with double bond at C2 and methyl substituent at C2. IUPAC Name: 2-methylbut-2-ene. **[1 Mark]**

• **Key Exam Result:** Alkene 'A' is 2-methylbut-2-ene: CH3-CH=C(CH3)2.

5. Explain the step-by-step mechanism of electrophilic nitration of benzene using a mixture of concentrated HNO3 and concentrated H2SO4. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Electrophile Generation):** H2SO4 acts as a Bronsted acid and protonates HNO3, which loses water to generate the electrophilic nitronium ion (NO2+): HNO3 + 2H2SO4 -> NO2+ + H3O+ + 2HSO4-. **[1 Mark]**

• **Step 2 (Arenium Ion Formation):** Electrophilic attack of NO2+ on the benzene pi-system generates a resonance-stabilized arenium ion (sigma-complex), temporarily disrupting aromaticity. **[1 Mark]**

• **Step 3 (Proton Abstraction):** Base HSO4- abstracts a proton from the sp3 carbon of the sigma-complex, restoring the stable 6 pi aromatic sextet and yielding nitrobenzene. **[1 Mark]**

• **Key Exam Result:** Nitration proceeds via nitronium ion generation, sigma-complex formation, and aromaticity restoration.

$$C_6H_6 + NO_2^+ \rightarrow [C_6H_6NO_2]^+ \xrightarrow{-H^+} C_6H_5NO_2$$

Electrophilic Aromatic Nitration Mechanism

6. Explain the following organic reactions with one balanced equation each: (a) Wurtz reaction, (b) Kolbe's electrolytic synthesis, (c) Friedel-Crafts alkylation of benzene. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Wurtz Reaction:** Alkyl halides react with sodium metal in dry ether to yield symmetrical higher alkanes: 2 CH3-Br + 2 Na -> CH3-CH3 + 2 NaBr (Ethane). **[1 Mark]**

• **(b) Kolbe's Electrolytic Synthesis:** Electrolysis of concentrated aqueous sodium acetate yields alkane at the anode: 2 CH3COONa + 2 H2O -> CH3-CH3 + 2 CO2 + H2 + 2 NaOH. **[1 Mark]**

• **(c) Friedel-Crafts Alkylation:** Benzene reacts with alkyl halide in presence of anhydrous AlCl3 Lewis acid to form alkylbenzene: C6H6 + CH3Cl -> C6H5-CH3 + HCl (Toluene). **[1 Mark]**

• **Key Exam Result:** Core synthetic pathways for alkane and substituted arene preparation.

7. How will you carry out the following organic conversions? (a) Ethyne to Benzene, (b) Benzene to Nitrobenzene, (c) Propene to 1-bromopropane. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Ethyne to Benzene:** Cyclic polymerization of ethyne by passing through a red-hot iron tube at 873 K: 3 CH#CH -> C6H6. **[1 Mark]**

• **(b) Benzene to Nitrobenzene:** Heating benzene with nitrating mixture (conc. HNO3 + conc. H2SO4) at 323-333 K: C6H6 + HNO3 -> C6H5NO2 + H2O. **[1 Mark]**

• **(c) Propene to 1-bromopropane:** Treating propene with HBr in the presence of benzoyl peroxide (Peroxide Anti-Markovnikov effect): CH3-CH=CH2 + HBr -> CH3-CH2-CH2-Br. **[1 Mark]**

• **Key Exam Result:** Standard board synthetic conversions involving cyclic polymerization, nitration, and radical addition.